

Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

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1 Model Structure and Assumptions

1.1 Health-State Definitions

Table 1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Definition	Utility	Monthly cost
Recurrence-free (RF)	No evidence of locoregional or distant recurrence following complete resection. Patients receive pembrolizumab (or combination) for up to 18 cycles.	0.83	\$0 (on-treatment costs captured separately)
Locoregional recurrence (LR)	Locoregional recurrence without distant metastases	0.64	\$3,000

1.2 State Occupancy Equations

State occupancy is determined by the partitioned survival model (PSM) approach:

$$\begin{aligned} \text{RF}(t) &= \text{RFS}(t) \\ \text{LR}(t) &= \text{DMFS}(t) - \text{RFS}(t) \\ \text{DM}(t) &= \text{OS}(t) - \text{DMFS}(t) \\ \text{Dead}(t) &= 1 - \text{OS}(t) \end{aligned}$$

These are **occupancy probabilities** derived from the three survival curves, not transition probabilities. The model does not simulate individual patient transitions; it computes the proportion of the cohort in each state at each time point directly from the survival functions.

The three survival functions are defined as follows:

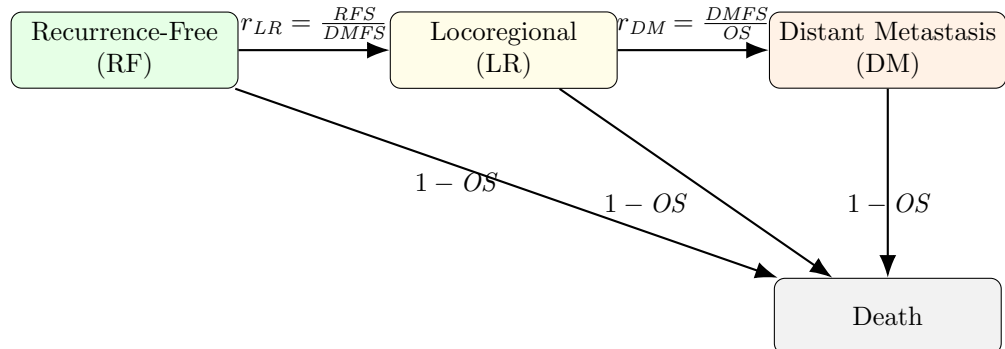
- **RFS (recurrence-free survival):** Time from randomization to first recurrence (locoregional or distant) or death from any cause. The RFS curve defines the proportion of patients who have not experienced any recurrence event.
- **DMFS (distant metastasis-free survival):** Time from randomization to first distant metastasis or death from any cause. The DMFS curve defines the proportion of patients who have not developed distant metastatic disease.
- **OS (overall survival):** Time from randomization to death from any cause.

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $\text{RF} = \text{RFS}$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $\text{LR} = \text{DMFS} - \text{RFS}$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $\text{DM} = \text{OS} - \text{DMFS}$.
- The dead state is the complement of overall survival: $\text{Dead} = 1 - \text{OS}$.

The constraint $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ is enforced at all time points to ensure non-negative state occupancy. This ordering is clinically meaningful: a patient cannot have distant metastasis without having had a recurrence event ($\text{RFS} \leq \text{DMFS}$), and a patient cannot die without having both events ($\text{DMFS} \leq \text{OS}$). The ratio-constrained structure $r_{\text{LR}} = \text{RFS}/\text{DMFS}$ and $r_{\text{DM}} = \text{DMFS}/\text{OS}$ guarantees this ordering by construction.

1.3 Model Schematic



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2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [?] using WebPlotDigitizer v4.7 (<https://automeris.io/WebPlotDigitizer>). Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS (60-month DMFS was not available from the published figure). The extracted values are reported in Table ??.

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Table 2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual point extraction with 5–10 points per curve per time point by a single reviewer. The 60-month landmark values (68.8% vs 49.1% for RFS; 92.2% vs 71.3% for OS) match the published landmark rates from the 5-year ASCO 2026 update [?], confirming the accuracy of the digitization process.

Comparison with published landmark rates: The published 5-year landmark rates (RFS: 68.8% vs 49.1%; OS: 92.2% vs 71.3%) are identical to the digitized values, indicating minimal extraction error. At earlier time points, the published 18-month RFS rates from the primary analysis (79% vs 62%) [?] differ from the digitized values (80.4% vs 62.2%) by 1.4 percentage points (combo) and 0.2 percentage points (pembro), consistent with typical digitization error for KM curves.

Availability of confidence intervals: The 5-year RFS landmark rates were reported with 95% confidence intervals: 68.8% (95% CI 56.3%–78.3%) for the combination arm and 49.1% (95% CI 33.3%–63.0%) for the pembrolizumab arm. These CIs were not incorporated into the primary analysis but are reported here for transparency. The parametric survival functions (Section ??) were fitted to the digitized point estimates using non-linear least squares, and the covariance of the fitted parameters was propagated through the PSA (Section ??).

subsectionParametric Model Fit

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captionDigitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm. RFS = recurrence-free survival; DMFS = distant metastasis-free survival; OS = overall survival. Digitized points are extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7.

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Table 3: Parametric Model Fit Comparison

Distribution	Parameters	AIC	RMSE	Accepted
<i>OS – Combo arm</i>				
Exponential	1	–	–	–
Weibull	2	–	–	–
Log-normal	2	-50.08	0.0045	Base case
Log-logistic	2	–	–	–
Generalized gamma	3	–	–	–
<i>OS – Pembro arm</i>				
[Full results from survival_model_selection.py]				
<i>r_{DM} – Combo arm</i>				
[Full results]				
<i>r_{DM} – Pembro arm</i>				
[Full results]				
<i>r_{LR} – Combo arm</i>				
[Full results]				
<i>r_{LR} – Pembro arm</i>				
[Full results]				

All 4,096 candidate distribution combinations (4 distributions for 6 curves, excluding over-fitted generalized gamma) satisfied the $RFS \leq DMFS \leq OS$ ordering constraint at all 480 monthly cycles.

2.2 Observed vs. Fitted Survival Curves

eginfigure[H]	[Figure S2: Observed vs. fitted RFS – Combo and Pembro arms]
eginfigure[H]	[Figure S3: Observed vs. fitted DMFS – Combo and Pembro arms]
eginfigure[H]	[Figure S4: Observed vs. fitted OS – Combo and Pembro arms]

3 Long-Term Validation

3.1 Long-Term Survival Diagnostics

Table 4: Long-Term Survival Predictions vs. General Population

Year	Combo OS	Pembro OS	GP Survival
5	0.916	0.741	0.945
10	0.825	0.516	0.891
15	0.693	0.392	0.808
20	0.561	0.284	0.704
30	0.214	0.109	0.463
40	0.000	0.000	0.000

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[Figure S5: OS vs. general population survival]

3.2 Structural Consistency Checks

Table 5: Structural Consistency Over 480 Monthly Cycles

Check	Result
RFS $\leq$ DMFS violations	0
DMFS $\leq$ OS violations	0
Negative state probabilities	0
Maximum state-sum deviation	$< 10^{-12}$
Clamp events (GP floor)	0

4 Model Inputs

4.1 Complete Parameter Table

Table 6: Complete Model Input Parameters

Parameter	Base	DSA Range	Source
<i>Survival parameters</i>			
OS μ (combo)	–	–	Fitted
OS σ (combo)	–	–	Fitted
OS μ (pembro)	–	–	Fitted
OS σ (pembro)	–	–	Fitted
r_{DM} μ (combo)	–	–	Fitted
r_{DM} σ (combo)	–	–	Fitted
r_{DM} μ (pembro)	–	–	Fitted
r_{DM} σ (pembro)	–	–	Fitted
r_{LR} μ (combo)	–	–	Fitted
r_{LR} σ (combo)	–	–	Fitted
r_{LR} μ (pembro)	–	–	Fitted
r_{LR} σ (pembro)	–	–	Fitted
<i>Costs (2026 USD)</i>			
Intismeran (per course)	\$200,000	–	Analyst estimate
Pembrolizumab (annual)	\$220,896	–	GoodRx
Sequencing (per course)	\$1,000	–	CMS
Administration (per cycle)	\$500	–	CMS
LR management (monthly)	\$3,000	–	Bensimon 2019
DM management (monthly)	\$12,000	–	Bensimon 2019
AE (incremental, one-time)	\$15,000	–	Bensimon 2019
<i>Utilities</i>			
RF state	0.83	0.03	Beta ($\alpha=129.3$, $\beta=26.5$)
Bensimon 2019, Johnston 2021			
LR state	0.64	0.03	Beta ($\alpha=163.2$, $\beta=91.8$)
Bensimon 2019, Johnston 2021			
DM state	0.55	0.03	Beta ($\alpha=150.7$, $\beta=123.3$)
Bensimon 2019, Zhang 2023			
AE disutility	0.05	–	Fixed (one-time)
Bensimon 2019, Zhang 2023			
<i>Model settings</i>			
Discount rate	3%	–	Standard
Time horizon	40 years	–	Lifetime
Cycle length	1 month	–	Standard
GP mortality	US 2019 Life Table	–	NVSS

Note: DSA ranges: RF 0.80–0.86, LR 0.61–0.67, DM 0.52–0.58 ($\pm 3\%$ of mean). Cost sources: Bensimon 2019, Zhang 2023.

5 Additional Results

5.1 Cost Decomposition

Table 7: Cost Decomposition by Component

Cost component	Combo	Pembro	Incremental
Intismeran	—	—	—
Pembrolizumab	—	—	—
Sequencing	—	—	—
AE/Administration	—	—	—
LR management	—	—	—
DM management	—	—	—
Total	—	—	—

5.2 QALY Decomposition

Table 8: QALY Decomposition by Health State

Health state	Combo	Pembro	Incremental
RF	—	—	—
LR	—	—	—
DM	—	—	—
Total	—	—	—

6 Structural Scenario Analyses

Table 9: Full Scenario Analysis Results

Scenario	ΔCost	ΔQALY	ICER	NMB @ \$150K	Source
Base case (GP-constrained)	—	—	—	—	—
Unconstrained (log-normal)	—	—	—	—	Methodological
Weibull OS	—	—	—	—	—
GP floor 48 months	—	—	—	—	—
GP floor 72 months	—	—	—	—	—
10-year horizon	—	—	—	—	—
20-year horizon	—	—	—	—	—
Waning v2 (5–10y)	—	—	—	—	—
Waning v2 (5–15y)	—	—	—	—	—
No direct OS benefit	—	—	—	—	—
Pricing: \$100K/course	—	—	—	—	—
Pricing: \$500K/course	—	—	—	—	—

7 Probabilistic Sensitivity Analysis

7.1 PSA Distributions and Covariance

Table 10: PSA Parameter Distributions

Parameter	Distribution	Source
Survival (μ, σ)	Bivariate normal (joint)	NLS covariance
Costs	Gamma (CV=0.2)	Standard
Utilities	Beta (SE=0.03)	Standard
Discount rate	Fixed	Standard

7.2 PSA Convergence

[Figure S6: PSA convergence – mean ICER at 100/500/1000/5000 iterations]

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7.3 Cost-Effectiveness Plane

[Figure S7: Cost-effectiveness plane (1,000 PSA iterations)]

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8 CHEERS 2022 Checklist

Table 11: CHEERS 2022 Health Economic Evaluation Reporting Checklist

#	Item	Reported	Section
1	Title	Yes	p.1
2	Abstract	Yes	p.2
3	Background/Objectives	Yes	Introduction
4	Health economic analysis plan	Yes	Methods
5	Study population	Yes	Methods
6	Setting and location	Yes	Methods
7	Comparators	Yes	Methods
8	Perspective	Yes	Methods
9	Time horizon	Yes	Methods
10	Discount rate	Yes	Methods
11	Selection of outcomes	Yes	Methods
12	Measurement of outcomes	Yes	Methods
13	Valuation of outcomes	Yes	Methods
14	Measurement and valuation of resources and costs	Yes	Methods
15	Currency, price date, and conversion	Yes	Methods
16	Rationale and description of model	Yes	Methods
17	Analytics and assumptions	Yes	Methods
18	Characterizing heterogeneity	N/A	Single population
19	Characterizing distributional effects	N/A	Single population
20	Characterizing uncertainty	Yes	Methods/Results
21	Approach to engagement with patients and others	N/A	Early CEA
22	Study parameters	Yes	Table 2 / Suppl. D
23	Summary of main results	Yes	Results
24	Effect of uncertainty	Yes	PSA/DSA
25	Effect of engagement with patients and others	N/A	Early CEA
26	Study findings, limitations, generalizability	Yes	Discussion
27	Source of funding	Yes	Title page
28	Conflicts of interest	Yes	Title page

References

- [] A. G. Bensimon, Z. Y. Zhou, M. Jenkins, et al. Cost-effectiveness of pembrolizumab for the adjuvant treatment of resected high-risk stage iii melanoma in the united states. *Journal of Medical Economics*, 22(10):981–993, 2019.
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- [] S. Zhang, A. G. Bensimon, R. Xu, et al. Cost-effectiveness analysis of pembrolizumab as an adjuvant treatment of resected stage iib or iic melanoma in the united states. *Advances in Therapy*, 40:3038–3055, 2023.
- [] A. Khattak et al. Five-year follow-up of keynote-942: individualized neoantigen therapy plus pembrolizumab. *Journal of Clinical Oncology*, 44:JCO2600835, 2026.
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